Inch-Pound

MIL-P-14852B (AR)
30 November 1988
SUPERSEDING
MIL-P-14852 A (MU)
18 April 1972

MILITARY SPECIFICATION

PROJECTILE, 105MM: ILLUMINATING, M314A3
METAL PARTS ASSEMBLY

This specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Projectile, 105MM: Illuminating, M314A3 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn: AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-M-6867	-	Magnetic Inspection Units
MIL-H-6875	-	Heat Treatment of Steels (Aircraft) Practice, Process For)
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410	-	Nondestructive Testing Personnel
MIL-STD-1167	-	Ammunition Data Card
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1949	-	Inspection Process, Magnetic Particle

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10534900	-	Projectile, 105MM, Illuminating M314A3 Metal Parts Assembly
8837635	-	Pallet for Fixed and Semi-Fixed Shell

INSPECTION EQUIPMENT DRAWINGS

8593429	-	Band Seating Testing Machine Indentor and Test Data
9280423	-	Magnetic Particle Standard for M314A3 Body
9280424	-	Magnetic Particle Standard for Base Plug, 105MM, M314A3

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DOCUMENTS

Operating Instructions for Use of Detroit Testing Machine for Band Seat Inspection

(Copies of specifications, standards, handbooks, drawings, documents and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Method E-8 - Tension Testing of Metallic Materials
ASTM Method E-10 - Brinell Hardness of Metallic Materials
ASTM Method B117 - Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, and referenced specifications.

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3.2 Heat treatment. Heat treatment applied to the projectile body and base plug shall be uniform throughout and shall be so conducted as to produce in each projectile body and base plug the mechanical properties specified by the applicable drawing. Metal forming involving plastic flow without subsequent heat treatment or stress relief is not permitted. Furnace equipment requirements and furnace calibration and test procedures shall be in accordance with the requirements of MIL-H-6875.

3.2.1 Hot forged - heat treatment method of manufacture. Components manufactured from forgings shall be subjected to heat treatment to attain the mechanical properties specified by the applicable drawing. Heat treatment shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit, followed by an oil quench. After quenching, the component forging shall be tempered without delay at a temperature above 750 degrees which will produce the required mechanical properties. After removal from the tempering furnace, the component forging shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees F, after which cooling may be accelerated by water spray.

3.2.2 Cold shaping and hot cup - cold draw method of manufacture. Cold working and subsequent stress relief shall be so accomplished as to produce in the projectile body the mechanical properties specified by the applicable drawing. Stress relief shall be accomplished at a temperature of 900 degrees F, minimum, following the final forming operations which result in plastic flow. The minimum time the projectile body shall be at the minimum stress relief temperature shall be one hour.

3.2.3 Tubing method of manufacturing. Bodies manufactured from tubing shall be heat-treated or stress relieved after forming of the nose.

3.2.4 Base plug. The base plug shall be heat-treated by normalizing or oil-quenching and tempering to obtain the specified mechanical properties in a transverse direction.

3.3 Mechanical properties.

3.3.1 Components. Mechanical properties specified on the applicable drawing shall apply throughout each component.

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3.3.2 Rotating band. After application and subsequent removal from the projectile body, the rotating band shall have the following mechanical properties:

Tensile strength - 30,000 pounds per square inch (psi) minimum
Elongation - 15 percent minimum in 1 inch

3.3.3 Base plug. Mechanical properties specified on the applicable drawing shall apply throughout the base plug.

3.4 Rotating band seating.

3.4.1 Band seat cleanliness. Prior to banding, the band seat shall be clean and free of oil, dirt, grease, rust or other foreign matter. Shot or sandblasting is not permitted for cleaning the band seat after knurling.

3.4.2 Application of band blank. The rotating band blank shall be pressed into the band seat, after final heat treatment of the projectile body. The band blank shall be applied in such a manner as not to distort the projectile body. The band blank must contact the sidewall of the band seat for a minimum of eight-tenths of the band seat depth measured from the outside surface of the projectile body. There shall be no gap between the band and the band seat sidewalls at the outside surface of the projectile body after the rotating band has been machined to final dimensions.

3.4.3 Rotating band clearance. The diametrical clearance between the rotating band and the band seat shall not exceed .006 inch, when the destructive method of testing is used; or .009 inch when the non-destructive method of testing is used.

3.4.4 Band bland - (gilding metal). The rotating band blank shall be in the form of a solid ring, annealed if necessary to comply with the applicable specification, except that when hot banding at not less than 800 degrees F, no prior annealing is necessary.

3.5 Metal defects

3.5.1 Body. The body, including the cavity, shall be free of cracks, splits, cold shuts, inclusions, pipe, porosity, bursts, folds, seams and other metal defects.

3.5.2 Body cavity. The body cavity shall be free from scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

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3.5.3 Base plug. The base plug shall be free from cracks, ›lits, cold shuts, inclusions, pipe, porosity, bursts, folds, ;ams and other metal defects.

3.5.4 Defect standards. Defect standards for metal defects .ll be established for each contract after the first month's ·oduction and must be approved by the technical engineering ;ency. However, the establishment of visual standards will not ·ive the requirement for compliance with 3.5.1, 3.5.2 and 3.5.3.

3.6 Proving ground functional tests. The projectile ;sembly shall reveal no evidence of metal parts failure when ›sted at excess pressure. Metal parts failure shall be:

a. Permanent deformation greater than .010 inch on either ıe body diameter or bourrelet diameter of the projectile.

b. Missing or loose rotating band.

c. Shear of rotating band as evidenced by complete lack of ıgraving.

d. Loss or breakup of metal parts components (except base lug).

3.7 Protective finish. Details of the proposed procedure or zinc phosphate coating to be used by the contractor shall be ubmitted in writing to the Government Technical Agency and ritten approval received prior to initiation of production. The roposed procedures shall include materials, processes and quipment to be used by the contractor, detailed methods of ontrol including limits for time, temperature, concentration and ll other information pertinent to the processes. Approval of rocess, materials, or equipment implies no guarantee of cceptance of the results obtained in use.

3.8 Process control of stress relief operation. A process ontrol document covering the stress relief operation shall be ubmitted to the technical agency through the contracting officer or review and approval prior to the initiation of production. his document shall include descriptions of procedures, quipment, process parameters, monitoring practices and alibration procedures and frequencies to be used for the stress elief operation. Data shall be included showing, based on emperature measurements, that the projectile bodies are at inimum temperature for the minimum time in accordance with 3.2.2 hen using the process parameters outlined in the process control

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document. The process control document shall be a part of the contractor's quality program plan. Revisions shall be submitted to the technical agency through the contracting officer for review and approval.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship. The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.10.1 Surface finish. The requirements for surface finish are as detailed on the applicable component drawing.

3.10.2 Protective coating. The protective coating shall be in accordance with the applicable drawing and the specification listed thereon.

3.10.3 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.10.4 Burr and sharp edges. No part shall have a burr or sharp edge which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.10.5 Foreign matter. No part or assembly shall contain dirt, grease, chips, oil, rust, corrosion or other foreign matter.

3.10.6 Painting. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with applicable specification.

3.10.7 Protection of rotating band The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities as indicated:

PART DESCRIPTION	DRAWINGS	QUANTITIES
Projectile Metal Parts Assembly Complete with Finished Base Plugs (unassembled)	10534900	20
Body Assembly, (Prior to painting)	10534901	4
Body, Projectile, (Prior to machining)	10534902	4
Body, Projectile, Machined, but Prior to Banding	10534902	4
Base Plug, Complete	10534903	6
Base Plug, Prior to Painting	10534903	4
Base Plug, Prior to Machining	10534903	3
Rotating Band Blank	8594851	5

Twenty-five of the above projectile metal parts assemblies and bases shall be forwarded to a designated load plant for loading and shall then be forwarded to a Government proving ground for ballistic testing. Ballistic testing will be performed subsequent to successful completion of inspections of Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

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Table I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-14852B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Projectile Metal Parts for Projectile Illuminating, M314A2		SEE BELOW
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Base Plug (Dwg 10534903)				
	Examination for defects	4		3.1	4.4.2.2
	Yield strength & percent elongation	2		3.1	4.5.1.2.2
	Salt spray test	2		3.1	4.5.3.2
	Macro etch	2		3.1	4.5.1.2.4
	Projectile Body (Dwg 10534902)				
	Examination for defects	4		3.1	4.4.2.4
	Yield strength & percent elongation	2		3.3	4.5.1.2.1
	Rotating Band Blank (Dwg 8594851)				
	Examination for defects	5		3.1	4.4.2.1
	Body Assembly (Dwg 10534901)				
	Examination for defects	4		3.1	4.4.2.5
	Projectile Metal Parts (Dwg 10534900)				
	Examination for defects	15		3.1	4.4.2.6
	Ballistic tests				
	a. Excess pressure	15		3.6	4.5.5
	Salt spray test	2		3.1	4.5.3.1

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectile shall contain:

a. Base plugs from not more than one forging manufacturer.

b. Projectile bodies from not more than one forging manufacturer.

4.4.1.2 Heat treatment lots. Heat treatment lots shall contain body forgings from not more than one heat of steel that has been heat treated in the same heat treating equipment, at the same temperature in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the mechanical properties test hereinafter prescribed.

4.4.1.3 Stress relieved lots. Stress relieved lots shall contain cold shaped bodies from not more than one heat of steel that have been cold shaped by the same process, utilizing the same tooling and equipment and are stress relieved at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.4 Tubing lots. Bodies manufactured from tubing shall be fabricated from tubing lots containing not more than one heat of steel. The bodies shall be nosed and heat-treated using the same equipment and procedures. Lots of bodies thus constituted shall be the basis for the selection of samples for the mechanical properties test hereinafter prescribed.

4.4.1.5 Base plug lots. A base plug lot shall consist of not more than 5,000 base plugs or fraction thereof produced by one manufacturer in one unchanged process in accordance with the same drawings, same drawing revisions, same specification and same specification revisions. The lot thus constituted shall be the basis for selecting the samples for the mechanical properties test hereinafter prescribed.

4.4.1.6 Miscellaneous lots. Small lots (such as bodies manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc.) may, at the discretion of the supplier, be grouped into miscellaneous heat treatment or stress relieved lots. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either 4.4.1.2, 4.4.1.3, 4.4.1.4 or 4.4.1.5 as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the mechanical properties test hereinafter prescribed.

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4.4.1.7 Use. The heat treatment, stress relieved, base plug or miscellaneous lot constituted per 4.4.1.2, 4.4.1.3, 4.4.1.4, 4.4.1.5 or 4.4.1.6 shall be the basis for selecting samples for mechanical properties tests. Projectile bodies or base plugs in each heat treatment lot shall be identified to that particular lot until satisfactory completion or mechanical properties tests.

4.4.1.8 Projectile lot. Heat treatment, stress relieved, base plug or miscellaneous lots that have been determined to have complied with the yield strength and percent elongation requirements of the applicable drawing may be regrouped for assembly into projectile lots.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-1235 or MIL-STD-105 as applicable.

b. Automatic gaging. Facilities equipped with Government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize capability in accordance with contract.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-14852B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Blank, Rotating, Band				8594851
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 10534901 / PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter		0.40%	3.1	Gage
102	Thickness		0.40%	3.1	Gage
103	Width		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Base Plug (prior to painting)		10534903
			NEXT HIGHER ASSEMBLY: 10534900

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Metal defective		100%	3.5.3	4.5.4
Major 101	Runout of force fit surface with large outside diameter		0.40%	3.1	Gage
102	Perpendicularity of front face of flange with force fit surface		0.40%	3.1	Gage
103	Diameter of force fit surface		0.40%	3.1	Gage
104	Diameter of large outside surface		0.40%	3.1	Gage
105	Depth of undercut, max.		0.40%	3.1	Gage
106	Length of force fit surface, min.		0.40%	3.1	Gage
107	Length total		0.40%	3.1	Gage
108	Thickness through base, min.		0.40%	3.1	Gage
109	Width of undercut, max.		0.40%	3.1	Visual
110	Finish on force fit surface improper	*		3.2	4.5.1.2.2
111	Yield strength and percent elongation	*		3.1	4.5.3.2
112	Protective coating (salt spray)	*		3.1	4.5.1.2.4
113	Macro etch				

NOTES: * Sample size as specified in applicable test method paragraph.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Base Plug (prior to painting)				10534903
					NEXT HIGHER ASSEMBLY
					10534900
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Angle at front of force fit surface		0.65%	3.1	Gage
202	Length from front face to shoulder		0.65%	3.1	Gage
203	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Base Plug (after painting)		10534903
			NEXT HIGHER ASSEMBLY 10534900

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Force fit surface damaged		0.40%	3.5.3	Visual
Minor 201	Paint on force fit surface		0.65%	3.1	Visual
202	Protective coating inadequate or damaged		0.65%	3.1	Visual
203	Wax emulsion missing from indicated surface		0.65%	3.1	Visual
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.4	Body (prior to banding)				10534902
					NEXT HIGHER ASSEMBLY 10534901
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.5.1	4.5.4.1
Major					
101	Diameter of rotating band seat		0.40%	3.1	Gage
102	Width of rotating band seat		0.40%	3.1	Gage
103	Ribs missing from band seat		0.40%	3.1	Visual
104	Scoring missing from band seat		0.40%	3.1	Visual
105	Evidence of sand or shot blast on band seat after scoring		0.40%	3.1	Visual
106	Runout of pitch diameter of thread with bourrelets and front face		0.40%	3.1	Gage
107	Runout of large cavity with bourrelets		0.40%	3.1	Gage
108	Runout of body with bourrelets		0.40%	3.1	Gage
109	Runout of base plug cavity with bourrelets		0.40%	3.1	Gage
110	Perpendicularity of face of base with bourrelets		0.40%	3.1	Gage
111	Diameter at front of large cavity, min.		0.40%	3.1	Gage
112	Diameter of small cavity, min. for min. length		0.40%	3.1	Gage
113	Diameter of large cavity		0.40%	3.1	Gage
114	Diameter of front bourrelet, min.		0.40%	3.1	Gage
115	Diameter of body (between rotating band and bourrelet)		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 10534902
4.4.2.4	Body (prior to banding)				NEXT HIGHER ASSEMBLY 10534900
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
116	Diameter of rear bourrelet, min.		0.40%	3.1	Gage
117	Diameter of chamfer, max. (at rear small cavity)		0.10%	3.1	Gage
118	Depth of small cavity, at front		0.40%	3.1	Gage
119	Length of taper section at front of large cavity, max.		0.40%	3.1	Gage
120	Thickness of wall, min. at small cavity		0.40%	3.1	Gage
121	Pitch diameter of thread, max.		0.40%	3.1	Gage
122	Pitch diameter of thread, min.		0.40%	3.1	Gage
123	Minor diameter of effective thread, max.		0.40%	3.1	Gage
124	Length of effective thread, min.		0.40%	3.1	Gage
125	Yield strength and percent elongation	*		3.3	4.5.1.2.1
Minor					
201	Location of band seat		0.65%	3.1	Gage
202	Foreign matter on band seat		0.65%	3.1	Visual
203	Diameter of nose outside		0.65%	3.1	Gage
204	Depth of large cavity		0.65%	3.1	Gage
205	Length to rear of forward bourrelet		0.65%	3.1	Gage
206	Length to ogive basic diameter		0.65%	3.1	Gage
207	Length total		0.65%	3.1	Gage
208	Diameter of base plug cavity		0.65%	3.1	Gage
209	Depth of base plug cavity		0.65%	3.1	Gage
210	Radii, chamfers or notches missing or incomplete		0.65%	3.1	Visual
211	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:
* Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Body Assembly (prior to painting)		10534901
			NEXT HIGHER ASSEMBLY 10534900

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Runout of rotating band seat with bourrelets		0.40%	3.1	Gage
102	Diameter of rotating band		0.40%	3.1	Gage
103	Length from rear face to start of rotating band		0.40%	3.1	Gage
104	Mechanical properties of rotating band	*		3.3.2	4.4.3.1.2.6
105	Rotating band and band seat clearance	*		3.4	4.4.3.2
Minor					
201	Diameter of rotating band undercut		0.65%	3.1	Gage
202	Length of rotating band flat		0.65%	3.1	Gage
203	Length of rotating band total		0.65%	3.1	Gage
204	Length of rotating band undercut		0.65%	3.1	Gage
205	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: * Sampling size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Metal Parts Assembly		10534900
			NEXT HIGHER ASSEMBLY: None

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Ballistic test (excess pressure)	*		3.6	4.5.5
Major					
101	Paint on thread		0.40%	3.1	Visual
102	Paint on walls of base plug cavity		0.40%	3.1	Visual
103	Rotating band damaged		0.40%	3.1	Visual
104	Wax emulsion missing from sidewall of base plug cavity		0.40%	3.1	Visual
105	Diameter of bourrelets, max.		0.40%	3.1	Gage
106	Salt spray	*		3.1	4.5.3.1
Minor					
201	Protective coating inadequate or damaged		0.65%	3.1	Visual
202	Marking missing, incorrect or illegible		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: * Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Pallet				88375835
					NEXT HIGHER ASSEMBLY: None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Pallet damaged (unserviceable)		0.65%	3.1	Visual
202	Sleeve missing		0.65%	3.1	Visual
203	Improperly packed		0.65%	3.1	Visual
204	Strapping missing, loose or broken		0.65%	3.1	Manual/Visual
205	Marking missing, incorrect or illegible		0.65%	3.1	Visual
206	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article Table, Quality Conformance Inspection Table and herein.

4.4.3.1 Mechanical properties.

4.4.3.1.1 Hardness.

4.4.3.1.1.1 Heat treatment lots. In order to determine uniformity of heat treatment, each body forging in the heat treatment lot (as defined by 4.4.1.2) shall be subjected to a hardness test with a Brinell hardness tester, using a 5-second minimum dwell or by any other method of hardness testing approved by the procuring activity.

4.4.3.1.1.2 Stress relieved lots. To assess uniformity of process, twenty-five cold shaped bodies from the first batch of each stress relieved lot (as defined by 4.4.1.3) processed through the stress relief furnace shall be subjected to a hardness test with a Brinell hardness tester or by any other method of hardness testing approved by the procuring activity.

4.4.3.1.1.3 Tubing lot. When mechanical properties of the tubing are altered by heat treatment or stress relief during the manufacture of components, the uniformity of process will be determined in accordance with 4.4.3.1.1.1 or 4.4.3.1.1.2 as applicable.

4.4.3.1.1.4 Miscellaneous lots. Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.4.3.1.1.1. Miscellaneous stress relieved lots shall be subjected to a hardness test in accordance with 4.4.3.1.1.2.

4.4.3.1.2 Tension test.

4.4.3.1.2.1 Heat treatment lots. Two projectile bodies, one of the hardest and one of the softest as determined by the hardness test, shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength of elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be taken from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

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4.4.3.1.2.2 Stress relieved lots. Two projectile bodies, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five projectiles shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot.

4.4.3.1.2.3 Tubing lots. Two projectile bodies, one of the hardest and one of the softest as determined by the applicable hardness test, shall be subjected to this test in accordance with 4.4.3.1.2.1.

4.4.3.1.2.4 Base plug lots. Two base plugs from each lot of base plugs (as defined by 4.4.1.5) shall be selected at random and subject to this test. Failure of any tension test specimen to comply with yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot.

4.4.3.1.2.5 Miscellaneous lots. Samples for tension test of miscellaneous heat treatment or stress relieved lots shall be selected in accordance with 4.4.3.1.2.1, 4.4.3.1.2.2 or 4.4.3.1.2.3 as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into a miscellaneous lot.

4.4.3.1.2.6 Macro Etch (base plug). Sample test quantities of base plugs shall be subject to a Macro Etch Test to comply with the cross forging requirement in accordance with 4.5.1.2.4.

4.4.3.1.2.7 Rotating band. One body assembly shall be selected from each 8-hour production or fraction thereof from each banding machine in operation and subject to this test to determine compliance with 3.3.2. Failure to comply shall be cause for rejection of the assemblies represented by the sample.

4.4.3.2 Band seating.

4.4.3.2.1 Tightness.

4.4.3.2.1.1 Nondestructive method. Five body assemblies from each 1-hour production per banding machine in operation shall be subjected to this test. Failure of one or more bands to comply with 3.4.3 shall be cause for rejection of the assemblies represented by the failing sample.

4.4.3.2.1.2 Destructive method. One body assembly from each 8-hour production or fraction thereof per banding machine in operation shall be subjected to this test to determine compliance with 3.4.3. Failure to comply shall be cause for rejection of the assemblies represented by the failing sample.

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4.4.3.2.1.3 Sidewall contact. Rotating bands of body assemblies selected under 4.4.3.1.2.7 or 4.4.3.2.1.2 shall be examined to determine compliance with 3.4.2. Failure to comply shall be cause for rejection of the assemblies represented by the sample.

4.4.3.3 Cleaning and application of protective coating. The sample size and acceptance criteria for inspection of the cleaning of the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.4.3.4 Proving ground functional tests. The following proving ground tests will be performed by the Government.

4.4.3.4.1 First article inspection. The quantities listed below shall be ballistically tested in the phase indicated as follows:

PHASE: Excess pressure QUANTITY: 25

4.4.3.4.2 Production sampling. Normal sampling shall be conducted until 3 consecutive lots (as determined in 4.4.3) have been accepted. Reduced sampling will then be instituted. Any failure occurring while on reduced sampling requires a return to normal sampling. Sample quantities for normal and reduced sampling on a per lot basis shall be as follows:

	NORMAL	REDUCED
PHASE: Excess pressure	5	2

4.4.3.4.3 Excess pressure. Samples provided in accordance with 4.4.3.4 will be subjected to this test. Failure of any projectile assembly to comply with 3.4 shall be cause for rejection of the sample.

4.4.3.5 Magnetic particle inspection (bodies and bases).

4.4.3.5.1 Projectile body. Heat treated projectile bodies after final machining but prior to banding shall be inspected by the magnetic particle inspection technique.

4.4.3.5.2 Base plugs. Heat treated base plugs after final machining but prior to applying protective coating shall be inspected by the magnetic particle inspection technique.

4.4.4 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval of inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

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4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification or calibration checks during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

4.4.4.2 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with MIL-STD-410. As a minimum, operating personnel shall be Level I and calibrating personnel shall be Level II.

4.5 Methods of inspection.

4.5.1 Mechanical properties.

4.5.1.1 Hardness. After final heat treatment but prior to machining, the Brinell hardness test shall be performed on the outside surface at the band seat location. The test shall be performed using the equipment and procedures for performing this test as prescribed by ASTM Method E-10, except that a 5-second minimum dwell shall be used. A direct reading hardness tester may be substituted for the Brinell scope (micrometer microscope) type, provided that the same size indenter ball and the same pressure is used and that 2000 body forgings will have been tested by both units to establish correlation. Continued correlation shall be assured by checking the hardness value of one body forging each hour by measuring the diameter of the indent with a Brinell scope, for comparison with the dial reading.

4.5.1.2 Yield strength and percent elongation.

4.5.1.2.1 Body. From the samples provided by the heat treat or stress relieved lots two tension test specimens conforming to the largest obtainable standard "R" type specimen, as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of the body at midwall, in such a manner that the center of the gage length of the machined test bar will ball in line coincident with the center of the rotating band seat. The number of specimens and their location are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the projectile body meet the drawing requirements. The criteria for performing this test shall be prescribed by ASTM Method E-8; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0) percent.

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4.5.1.2.2 Base plug. From the samples provided by 4.4.3.1.2.3, one tangentially transverse tension test specimen conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken from each base plug. The centerline of the specimen shall be between .50 and .75 inches from the centerline of the base plug. The number of specimens and their location are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties meet the drawing requirements. The criteria for performing this test shall be as prescribed by ASTM Method E-8, except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0).

4.5.1.2.3 Rotating band (gilding metal). The samples provided by 4.4.3.1.2.6 shall have the rotating band removed and carefully flattened without hammering so as to hold to a minimum any additional cold working. The specimen blank shall be machined so that all serration marks from the band seat are removed. A test specimen having a one inch gage length and a one-half inch gage width, and grip ends of sufficient length to accommodate the tensile machine shall be removed from the blank and tested in accordance with the procedures prescribed by ASTM Method E-8.

4.5.1.2.4 Macro Etch (base plug). One base plug from each standard heat treatment lot and one from each miscellaneous heat treatment lot shall be subjected to a Macro Etch Test. The base plug shall be sectioned through its diameter and one of the sectioned surfaces prepared by grinding the surface to a flat. This part shall then be immersed in a macro etch solution of 50 percent hydrochloric acid and 50 percent water and heated to a temperature of 180 degrees F, minimum. The base part shall remain in the macro etch solution until flow lines become clearly visible. The part shall then be rinsed with water, dried and visually examined for compliance with Dwg. 10534903.

4.5.2 Band seating.

4.5.2.1 Nondestructive method. Rotating bands of the sample provided in accordance with 4.4.3.2.1.2 need not be processed through the band machining line, although the band should have a light surface cut (.020 max.) to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on 2 points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diameterical measurements per band; these points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body

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and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat, the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

4.5.2.3 Sidewall contact. Samples provided by 4.4.3.2.1.3 shall be visually examined for contact with the sidewalls of the band seat.

4.5.3 Protective coating. A salt spray test shall be performed on the projectile body and base plug utilizing the equipment, methods and procedures specified in ASTM B-117 and in accordance with the applicable drawings and specifications. Failure to comply will be cause for rejection of the production represented by the sample.

4.5.3.1 Projectile body. Salt spray testing shall be conducted on production projectile bodies after the protective coatings have been applied (see Dwg. 10534900). Sampling shall be in accordance with TT-C-490 when on reduced sampling (twice per week) one sample shall be taken during the first 30 minutes of production and the second taken during the last day of production that week.

4.5.3.2 Base plug. Salt spray testing shall be performed on production base plugs after the protective coating has been applied (see Dwg. 10534903). Sampling shall be as specified in TT-C-490. When on reduced sampling (twice per week), one sample during the first 30 minutes of production and the second taken during the last day of production that week.

4.5.4 Magnetic particle inspection.

4.5.4.1 Projectile body. Heat treated or stress relieved projectile bodies after final machining but prior to banding shall be inspected by the magnetic particle inspection technique. The procedures for performing this test shall be as specified in MIL-STD-1949 for wet flourescent method with indirect magnetization by the continuous method or the residual method. The entire body inside and outside shall be inspected after inducing a circular field. The magnetic particle standard per Dwg. 9280423 shall be utilized to establish inspection sensitivity. Any body which fails to comply with the requirements of 3.5 shall be rejected. Projectile bodies which comply shall be stamped with the letter "M" as specified on the applicable drawing.

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4.5.4.2 Base plug. Base plugs after final machining shall be inspected by the magnetic particle inspection technique. The procedures for performing this test shall be as specified in MIL-STD-1949 for wet flourescent method with indirect magnetization by the continuous or residual method. The magnetic particle standard per Dwg. 9280424 shall be utilized to establish inspection sensitivity. Any base plug which fails to comply with the requirements of 3.5 shall be rejected. Base plugs which comply shall be stamped with the letter "M" as specified on the applicable drawing.

4.5.4.3 Suspension. The solid content of the suspension shall be maintained at the level recommended by the suspensoid manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by Section 4 of MIL-STD-1949. When performing this test for solid content of the suspension, if it is determined that it does not comply, all bodies that have been tested and accepted since the last satisfactory test was performed shall be retested after the suspension has been brought to the proper level. The magnetic particle test shall be performed on equipment that complies with the requirements of specification MIL-M-6867.

4.5.5 Proving ground function tests.

4.5.5.1 Excess pressure. The metal parts assemblies supplied in accordance with 4.4.3.4 inert loaded and assembled in a complete round, will be fired for recovery in a gun for which the projectile is standard, the tube of which will have a minimum of 25 percent life remaining based on current gun tube condemnation criteria. A propellant charge adjusted to produce 109 to 117 percent of the upper pressure limit of the propellant shall be used with the applicable projectile. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.6. Recovered projectiles will be examined to determine compliance with 3.6.

5. PACKAGING

5.1 Packing and marking. Level C - Packing and marking shall comply with applicable drawings and specifications, MIL-STD-1169 and the following:

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5.2 Packing.

5.2.1 Body assembly. Body assemblies, part No. 10534901, shall be palletized in accordance with Dwg. 8837835.

5.2.2 Base plugs. Base plugs, part No. 10534903, shall be packed in fiberboard shipping containers in accordance with the requirements of MIL-STD-1169. Pack twenty (20) plugs in a 4 X 5 pattern, using interlocking separators between cells.

5.3 Marking.

5.3.1 Body assembly. Palletized body assemblies shall be marked in accordance with Dwg. 8837835.

5.3.2 Base plug. Packaged base plugs shall be marked in accordance with requirements of MIL-STD-1169.

5.4 Shipping. When component parts from more than one lot are shipped as a carload, each lot must be kept separated and division between lots clearly indicated so as to prevent mixing in transit.

5.5 Data cards. The contractor shall prepare and submit data cards, in the number to be determined by the contracting officer, in accordance with MIL-STD-1167.

6. NOTES

6.1 Intended use. The projectile metal parts contained in this specification are intended for use with Projectile, 105MM, Illuminating: M314A3.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, USAMCCOM (ARDEC), ATTN: AMSMC-QAR-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirement List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of

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DOD FAR Supplement Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph	Data Requirement Title	Applicable DID
3.2, 3.4.4, 3.7	Procedure Report	DI-T-5372*
3.9	Process Control Document	DI-R-5297A
4.4.4	Inspection Equipment Description Documentation	DI-R-10054

*Delete a, b, c, d of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits.

6.5 Submission of ballistic test data and data cards. In addition to the normal distribution, one copy of all ballistic data and ammunition cards shall be forwarded to Commander, USAMCCOM, (ARDEC), ATTN: AMSMC-QAR-Q, Picatinny Arsenal, NJ 07806-5000.

6.6 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.7 Band seating. To obtain the operating procedures for use of the Detroit Testing Machine (prepared by Inspection Engineering Office, US Army, Frankford Arsenal), submit requests to Commander, USAMCCOM (ARDEC), ATTN: AMSMC-QAR-I, Picatinny Arsenal, NJ 07806-5000

6.8 Steel blooms, billets, slabs and bars.

6.8.1 Steel grade. It is the responsibility of the manufacturer of the projectile metal parts to select a steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machinist to produce the projectile to the dimensions, mechanical properties and metal integrity specified as an end item requirement.

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6.8.2 Raw material. The supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade, and type with dimension, weight, soundness and freedom from the standard AISI grade selected by the projectile manufacturer and complying with the specified ASTM designation and tests selected by the projectile manufacturer.

6.8.3 Forging quality processing techniques. In the application of carbon steel, blooms, billets, slabs and bars, forging quality should be applied with the expectation that there will be variables in the metal cross-sectional dimensions, in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machinist have been engineered in terms of equipment sequence of operations, tooling designs, handling devices, and processing techniques having the capabilities of compensating for such variables.

6.9 Preproduction sample/hardness range (projectile bodies and base plugs). To minimize the need for reheat treating material or for having miscellaneous heat treatment lots it is recommended that the contractor process a preproduction sample to establish what should be an acceptable hardness range for each mill heat. This hardness range could then be used for screening bodies during processing of the heat treatment lot from that mill in accordance with paragraph 4.5.1.2.

6.10 Ballistic testing. Any evidence of excessive deformation in any phase of the proving ground tests will result in an immediate investigation by AMCCOM personnel.

6.11 Subject term (keyword) listing.

Ammunition
Artillery
Illuminating
Military Specification
Munition
Projectile, 105MM: Illuminating, M314A1, Metal Parts Assembly

6.12 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1315-A981)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-14852B (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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INCH-POUND

MIL-P-14852B (AR)
AMENDMENT 1
22 June 1994

MILITARY SPECIFICATION

PROJECTILE, 105MM: ILLUMINATING,
M314A3, METAL PARTS ASSEMBLY

This amendment forms a part of MIL-P-14852B (AR), dated 30 November 1988, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

3.4, Delete paragraph in its entirety and substitute with:

"3.4 Rotating band seating.

3.4.1 Rotating band cleanliness. Prior to banding, the band seat shall be clean and free of oil, dirt, grease, rust or other foreign matter. Shot or sand blasting is not permitted for cleaning the band seat after knurling.

3.4.2 Band blank - (gilding metal). The rotating band blank shall be in the form of a solid ring and shall be annealed to comply with the applicable specification. Annealing prior to banding shall not be necessary in the case of hot banding in excess of 800 degrees F.

3.4.3 Application of band blank. The rotating band blank shall not be pressed into the band seat after final heat treatment of the projectile body. The band blank shall be applied in such a manner as not to distort the projectile body.

3.4.4 Sidewall contact. The band blank must contact the sidewall of the band seat for a minimum of eight-tenths of the band seat depth, measured from the outside surface of the projectile body to the rotating band undercut surface.

3.4.5 Sidewall gap. There shall be no gap between the rotating band and the band seat sidewalls at the outside surface of the projectile body after the rotating band has been machined to final dimensions.

3.4.6 Rotating band clearance. The diametrical clearance between the the rotating band and the band seat shall not exceed .006 inch, when the destructive method of testing is used, or .009 inch when the non-destructive method of testing is used."

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3.6, Delete paragraph in its entirety and substitute the following:

"3.6 Proving ground functional tests. The projectile assembly shall reveal no evidence of failure when ballistically tested at excess pressure (see 4.5.5.1). Metal part failure shall be defined as a breakup either in the bore of the weapon or in flight."

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4.4.2.4, Add the following new requirement paragraph to Major category 105:

"3.4.1"

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4.4.2.5: Change Major categoy 104 from "4.4.3.1.2.6" to "4.4.3.1.2.7". Add the following new major defect 106:

"106 Sidewall gap .40% 3.4.5 Gage"

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4.4.3.1.2.7, Delete in its entirety and substitute the following:

"4.4.3.1.2.7 Rotating band. The sample of rotating band for testing in accordance with 4.5.1.2.3 for compliance with 3.3.2 shall be provided as part of the sample taken from 4.4.3.2.1.3. The sample size, frequency and rejection criteria is the same as that described in 4.4.3.2.1.3."

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4.4.3.2.1.3, Delete in its entirety and replace with:

"4.4.3.2.1.3 Sidewall contact. One body assembly shall be selected at the beginning of each shift (not to exceed ten hours) from each shift (not to exceed ten hours) from each banding machine and sectioned in accordance with 4.5.2.3 for compliance with 3.4.4 and also 3.3.2 (see 4.4.3.1.2.7). If, for any reason, there is a change in operator, machinery, or material (by lot number) during a shift, another sample shall be taken. Failure to comply with the requirements of the specification shall be cause for rejection of all material processed from the previous sample."

4.5.1.2.3, Change reference paragraph from "4.4.3.1.2.6" to "4.4.3.1.2.7".

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4.5.2, Delete paragraph in its entirety and replace with:

"4.5.2 Rotating band seating.

4.5.2.1 Clearance (Nondestructive method). The samples provided by 4.4.3.2.1.1 shall be tested for the requirements in 3.4.6 utilizing the equipment andprocedures prescribed in the Detroit Testing Machine Manual (see 6.7).

4.5.2.2 Clearance (Destructive method). The body assembly selected in accordance with 4.4.3.2.1.2 for compliance with 3.4.6 need not be processed through the band machining line, although the band should have a light surface cut (.020" max.) to provide a uniform surface for measurement. The diameter of each band shall be measured at three locations, sixty degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This results in six (three pairs of two) diametrical measurements per band. The points of measurement shall be marked on the band and the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The band shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of twelve measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the two corresponding two thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

4.5.2.3 Sidewall contact. The samples provided in accordance with 4.4.3.2.1.3 shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other section shall be sectioned into two parts, exposing four surfaces for the viewing of the band swaged into the band seat. The exposed surfaces shall be ground, and the metal chip shall be removed prior to measurement for depth of sidewall contact to determine compliance with 3.4.4."

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4.5.5.1, Change the second sentence to the following:

"A propellant charge adjusted to yield a mean pressure of 42,500 psi with no maximum individual exceeding 44,400 psi shall be used with the applicable projectile."

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Change the last sentence of 4.5.5.1 to the following:

"Recovery and examination of metal parts for any abnormal condition shall be accomplished for information only."

Custodian:
Army-AR

Preparing activity:
Army-AR

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